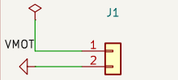
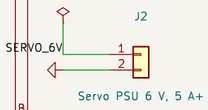
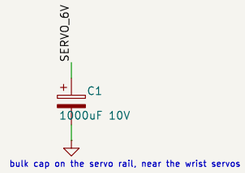
POWER



Motor PSU 12–24 V

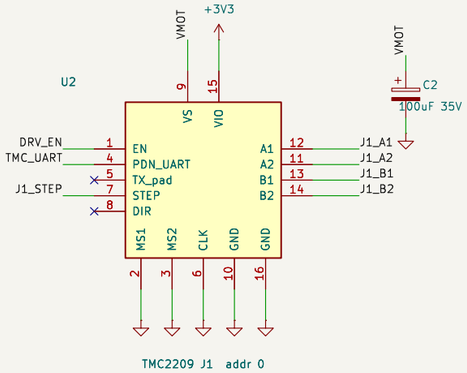


Servo PSU 6 V, 5 A+

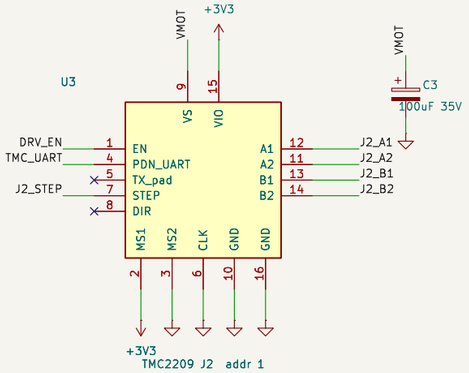


bulk cap on the servo rail, near the wrist servos

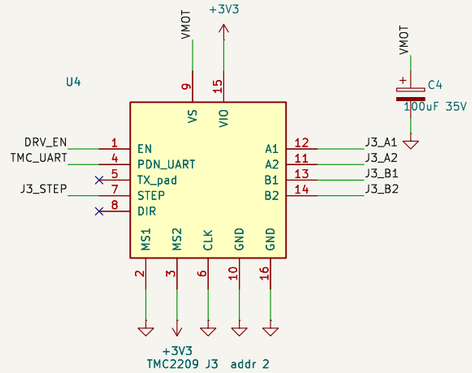
STEPPER JOINTS (TMC2209 on one UART bus, DIR unused)



TMC2209 J1 addr 0

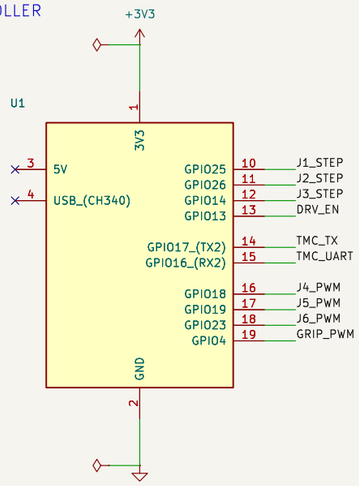


TMC2209 J2 addr 1



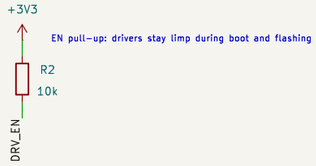
TMC2209 J3 addr 2

CONTROLLER



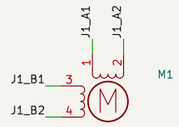
ESP32-DevKit (WROOM-32, CH340)

USB to the PC: power + serial 115200 baud

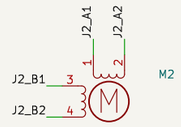


EN pull-up: drivers stay limp during boot and flashing

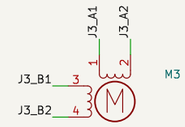
half-duplex UART: the only 1k is in TX



17HS4401S (J1)

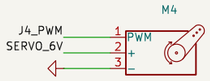


17HS4401S (J2)



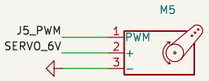
17HS4401S (J3)

SERVO JOINTS + GRIPPER (LEDC PWM 50 Hz, own 6 V supply)



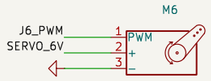
MG996R (J4)

270 deg, wrist pitch



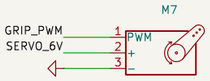
MG995 (J5)

270 deg, wrist roll



MG995 (J6)

270 deg, wrist pitch 2



MG995 360 (GRIP)

CONTINUOUS rotation, tendon spool

WIRING NOTES (what bit us, roughly in order of how much time it cost)

1. TMC2209 UART needs VS. VIO only powers the I/O ring; with no motor supply the driver never answers.
2. R7 on each StepStick bridged pad 2–3 so PDN_UART reaches header pin 4. Check: pin 4 to GND reads ~20k.
3. One 1k total, in the ESP32 TX line (GPIO17). RX (GPIO16) goes straight to the bus. One wire, half duplex.
4. Bus address latched from MS1/MS2 at power-up: J1=0 (GND,GND), J2=1 (3V3,GND), J3=2 (GND,3V3). Max 4 per bus.
5. DIR is NOT wired: the DIR inputs on these modules are damaged. Direction = GCONF.shaft over UART (fw 1.2+).
6. EN is active low and shared. The 10k pull-up keeps the motors limp while the ESP32 boots or reflashes.
7. CLK to GND = internal oscillator. Microstepping is set over UART (MRES); MS1/MS2 are address pins only.
8. Never plug or unplug a motor while VMOT is live — the back-EMF spike kills the driver.
9. Servos never take power from the dev board. Common GND between ESP32, motor PSU and servo PSU is mandatory.
10. GRIP is a continuous-rotation servo: pulse width = speed, ~1500 us = stop. See docs/calibration.md.
11. 17HS4401S coils: usually black/green = A, red/blue = B. Confirm with a meter (a few ohms within a coil).

Pin numbers read from the controller (pins command), not from older docs
As built: breadboard / point-to-point. Generated by tools/make_wiring_schematic.py

Samer Haddadin

Sheet: /
File: arm-wiring.kicad_sch

Title: Robotic arm — wiring as built

Size: A3	Date: 2026-11-07	**Rev: fw 1.3.0**
KiCad E.D.A. 9.0.7		Id: 1/1